Hyperspectral Anomaly Detection Based on Tensor Deep Subspace Prior Modeling

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***Abstract*—Hyperspectral anomaly detection (HAD) can be formulated as a binary classification problem, where the accurate separation of anomalies from the background relies on the precise features extraction or the effective prior representation of the background. Traditional methods based on hand-crafted regularizations, typically integrating low-rank and local smoothness priors of hyperspectral background images, offer strong interpretability. However, these methods rely on predefined regularization forms, which limit their ability to provide fine-grained, data-adaptive prior modeling for each specific instance. To overcome this limitation, we propose a novel model termed implicit Tucker subspace representation with sparsity regularization (ITSR-ℓ_1). The motivation for ITSR-ℓ_1 arises from the following three key points: 1) the low-rank Tucker decomposition can explicitly model the inherent low-rank structure of hyperspectral background images; 2) the core tensor obtained through Tucker decomposition tends to exhibit significant sparsity, which can be exploited further to enhance the modeling of the low-rank background component; and 3) the local smoothness of the original image can be transferred to the continuity of the column vectors of the factor matrices derived from Tucker decomposition. By parameterizing these vectors with implicit neural networks, ITSR-ℓ_1 enables data-adaptive, fine-grained characterization of continuity priors across the background.**

***Index Terms*—Data-adaptive prior modeling, hyperspectral anomaly detection (HAD), implicit Tucker subspace representation with sparsity regularization (ITSR-ℓ_1), low-rank Tucker decomposition.**

NOMENCLATURE

x	Scalar.
$\mathbf{x}$	Vector.
$\mathbf{X}$	Matrix.
$\boldsymbol{\mathcal{X}}$	Tensor.
$\boldsymbol{\mathcal{X}}_{(i,j,k)}$	(i,j,k)th element of tensor $\boldsymbol{\mathcal{X}}$.
$\texttt{unfold}_i(\cdot)$	Tensor unfolding along the i-mode.
$\boldsymbol{\mathcal{X}}^{(i)}$	Mode-i unfolding matrix, $\boldsymbol{\mathcal{X}}^{(i)} = \texttt{unfold}_i(\boldsymbol{\mathcal{X}})$.
$\texttt{fold}_i(\cdot)$	Folding operator, the inverse of $\texttt{unfold}_i(\cdot)$.
$\|\boldsymbol{\mathcal{X}}\|_F$	Tensor Forbenius norm, $\left(\sum_{i,j,k} \boldsymbol{\mathcal{X}}^2_{(i,j,k)}\right)^{1/2}$.
$\|\boldsymbol{\mathcal{X}}\|_{\ell_1}$	Tensor ℓ_1-norm, $\sum_{i,j,k} \|\boldsymbol{\mathcal{X}}_{(i,j,k)}\|$.

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I. INTRODUCTION

HYPERSPECTRAL images (HSIs) consist of hundreds of continuous and finely divided spectral bands, from which the spectral curve at any spatial point can be obtained by extracting pixel values across all bands. This allows HSIs to contain both visual information and rich spectral information, making them widely used in fields, such as camouflage recognition, military reconnaissance, deep space exploration, and pollution detection. Hyperspectral anomaly detection (HAD) identifies anomalous pixels based on the differences between the spectral curves of anomalies and background pixels, but the intrinsic limitations of HSIs and the lack of prior information make accurate detection challenging. To address this, prior information for HSIs is generally characterized using either data-driven [1], [2], [3], [4], [5], [6], [7], [8], [9], [10], [11], [12] or model-driven [13], [14], [15], [16], [17], [18], [19], [20], [21], [22], [23], [24], [25], [26], [27], [28], [29], [30] approaches. Both strategies are commonly employed in HAD and have attracted significant attention.

Data-based methods have been extensively applied to HAD, often incorporating deep learning either partially or fully. Some approaches utilize supervised neural networks, trained with paired datasets, to infer background or anomaly structures directly from the observed input data [31], [32]. Although such methods eliminate the need for prior knowledge, they often suffer from a reliance on labeled samples and are prone to model mismatch. To overcome the dependency on paired data, the unsupervised and semi-supervised techniques based on autoencoders (AEs) [1], [2], [3], [4], [5] and generative adversarial networks (GANs) [6], [7], [8], [9], [10], [11] have been proposed to reconstruct only the background. Since anomalies are typically unreconstructable, they manifest as large residuals, which can be used for detection either directly or after postprocessing. For instance, some powerful network architectures can further characterize the hyperspectral data and highlight background and anomalous differences

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[33], [34], [35], [36], [37], [38], [39]. Blind-spot networks leverage the distinct spectral characteristics of anomalous targets, producing higher reconstruction errors for anomalies and lower errors for background pixels, thereby enhancing anomaly localization [33], [34]. In addition, some studies have introduced Mamba, which integrates state-space models with deep learning to effectively capture long-range spatial–spectral dependencies, enabling feature extraction from local to global perspectives [35], [36]. Meanwhile, considering the strong performance of the Kolmogorov–Arnold network (KAN) in nonlinear feature mining and representation, the AD method based on KAN [37] achieves stronger background feature extraction and characterization capabilities than traditional network architectures, such as MLP and CNN. Given the powerful generative modeling capabilities of diffusion models for HSI [38], [39], [40], some HAD methods based on diffusion models achieve background reconstruction or accurate background estimation by learning the latent background distribution, thereby erasing the background [39] or enhancing the subsequent detection capabilities of the anomaly masks [38]. Furthermore, neural networks can assist anomaly detection in auxiliary roles. Deep features extracted by pretrained networks have been utilized to design more effective detectors [16], [41], [42], [43]. Despite their effectiveness, these methods still face challenges in generalization and in the integration of prior information specific to hyperspectral data.

Unlike data-based methods that learn direct mappings, model-driven approaches incorporate prior knowledge through carefully designed regularization terms. Over time, the prior assumptions in HAD have evolved from simple statistical models to more structured representations. These priors can be broadly categorized into three types: Gaussian modeling, low-rank representation, and local smoothness characterization. Gaussian-based models include Reed–Xiaoli (RX) and its variants [13], [14], [15]. These approaches are based on the assumption that the background conforms to a predefined statistical distribution, which often fails in practical scenarios, thereby limiting detection accuracy. Moreover, this assumption is inadequate for capturing the complex spectral correlations that are intrinsic to hyperspectral data. Low-rank matrix/tensor decomposition, such as robust principal component analysis (RPCA) [19] and tensor RPCA (TRPCA) [44], that decomposes HSIs into low-rank background and sparse anomaly [19], [21]. CP decomposition [20], Tucker decomposition [41], and tensor ring [45] have been widely used to model the background in HSI by factorizing them into low-rank structures, which can effectively describe the global spectral relationships. The advanced models like multiscale entanglement renormalization ansatz (MERA) further enhance representation efficiency in high-dimensional data [46]. In addition to low-rankness, smoothness is commonly used in HAD due to its capability to model spatial and spectral smoothness for the background and improve the anomaly detection performance by preserving edges while suppressing noise [23], [25], [26]. Although handcrafted TV regularization possesses strong interpretability, such that methods predefine the form and weighting of the regularization term. Specifically, TV regularization assumes a uniform degree of continuity across all points, which is clearly inconsistent with real-world data. As a result, manually designed regularization fails to provide a data-adaptive and fine-grained prior modeling mechanism tailored to each data point requiring restoration.

To address the limitations of the existing low-rank background modeling methods, we introduce a new framework termed implicit Tucker subspace representation (ITSR). The key idea is to integrate Tucker decomposition with implicit neural representations (INRs) for more flexible and data-adaptive prior modeling. First, instead of directly applying Tucker decomposition in a purely linear form, we parameterize its factor matrices using coordinate-based INRs. This implicit parameterization allows the model to capture spatial–spectral continuity in a fine-grained and data-driven manner, overcoming the rigidity of hand-crafted smoothness priors, such as TV. Second, although the decomposition form differs from conventional Tucker models, we consistently observe that the resulting core tensor still exhibits strong sparsity in its element distribution—a property closely tied to the intrinsic low-rank structure of hyperspectral backgrounds. Motivated by this observation, we further incorporate an ℓ_1 penalty on the core tensor to explicitly exploit its sparsity and enhance background representation accuracy. Unlike previous methods that exploit core-tensor sparsity in the gradient domain [41], ITSR leverages INR to directly encode the continuity structure of the original data. This allows us to impose sparsity on the core tensor in the original domain, leading to a more accurate characterization of the background low-rank property. By combining INR-parameterized factor matrices with sparsity-enhanced core tensor modeling, we develop the complete method ITSR-ℓ_1, which achieves adaptive, fine-grained, and more expressive low-rank prior modeling for HAD. The contributions of this research are articulated as follows.

1) The low-rank Tucker decomposition explicitly captures the inherent low-rank structure of the background; the core tensor obtained through the Tucker decomposition further exhibits significant sparsity; here, we utilize the ℓ_1 norm to enhance the modeling of background low-rank properties.
2) The continuous structure observed in the background factor matrices derived from the Tucker decomposition indicates a local smoothness of the background, using implicit neural networks to parameterize these factor vectors, which enables the proposed method to achieve data-adaptive, fine-grained characterization of background.
3) Extensive experiments on four real-world datasets, evaluated using 3-D ROC curves and visual analysis, convincingly demonstrate the superior performance of the proposed method over current state-of-the-art techniques.

II. Methodology

A. Preliminaries

This section introduces the notations summarized in Nomenclature and presents several important definitions that form the basis for the subsequent sections.

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Definition 1 (Tucker Rank [21]): The Tucker rank of a third-order tensor $\mathcal{X} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ is defined as

$$\text{rank}_T(\mathcal{X}) := \left(\text{rank}\left(\mathbf{X}_{(1)}\right), \text{rank}\left(\mathbf{X}_{(2)}\right), \text{rank}\left(\mathbf{X}_{(3)}\right)\right) \quad (1)$$

the notation can be simplified to $(\text{rank}_T(\mathcal{X}))_{(i)} := \text{rank}(\mathbf{X}_{(i)})$.

Definition 2 (Tucker Decomposition [21]): Given a tensor $\mathcal{X} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ with Tucker rank $\text{rank}_T(\mathcal{X}) = (r_1, r_2, r_3)$, it can be expressed via a Tucker decomposition as

$$\mathcal{X} = [\mathcal{G}; \mathbf{U}, \mathbf{V}, \mathbf{W}] = \mathcal{G} \times_1 \mathbf{U} \times_2 \mathbf{V} \times_3 \mathbf{W} \quad (2)$$

where the core tensor $\mathcal{G} \in \mathbb{R}^{r_1 \times r_2 \times r_3}$, and the factor matrices are $\mathbf{U} \in \mathbb{R}^{n_1 \times r_1}$, $\mathbf{V} \in \mathbb{R}^{n_2 \times r_2}$, and $\mathbf{W} \in \mathbb{R}^{n_3 \times r_3}$.

Definition 3 (INR [47]): An INR is defined as a continuous function parameterized by a neural network with a sine activation function

$$f_\theta : \mathbf{x} \in \mathbb{R}^d \longrightarrow \mathbf{y} \in \mathbb{R}^k \quad (3)$$

where $\mathbf{x}$ denotes an input coordinate, $\mathbf{y}$ is the corresponding signal value, and d denotes the dimensions of the input; for example, $d = 1$ for a 1-D signal and $d = 2$ for an image since the image has two dimensions. The function f_θ is typically implemented as a multilayer perceptron (MLP) with parameters θ, enabling continuous and differentiable modeling of complex signals.

Definition 4 (STV [48]): The spatial total variation (STV) regularization term of tensor $\mathcal{X}$ is defined as

$$\|\mathcal{X}\|_{\text{STV}} = \|\nabla_1 \mathcal{X}\|_1 + \|\nabla_2 \mathcal{X}\|_1 \quad (4)$$

where the operators ∇_1 and ∇_2 represent the first-order finite differences along the horizontal and vertical axes of the background, respectively, and $\|\cdot\|_1$ refers to the tensor ℓ_1-norm.

B. Problem Formulation

In HAD, the goal is to accurately separate anomalies from the background, which relies on precise background reconstruction. An HSI, represented as a third-order tensor $\mathcal{Y} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$, can be decomposed as

$$\mathcal{Y} = \mathcal{B} + \mathcal{E} + \mathcal{N} \quad (5)$$

where $\mathcal{B}$, $\mathcal{E}$, and $\mathcal{N}$ denote the background, anomalies, and noise, respectively. The dimensions n_1, n_2, and n_3 correspond to the spatial height, spatial width, and number of spectral bands, respectively.

Tucker decomposition is a classical and widely adopted tensor decomposition technique, which is particularly effective for capturing the intrinsic low-rank structures of backgrounds in multidimensional data. It represents a tensor through a compact core tensor together with a set of modewise subspace bases with Tucker rank $\mathbf{r} = (r_1, r_2, r_3)$, as formulated by

$$\mathcal{B} = \mathcal{G} \times_1 \mathbf{U} \times_2 \mathbf{V} \times_3 \mathbf{W} \quad (6)$$

where $\mathcal{G} \in \mathbb{R}^{r_1 \times r_2 \times r_3}$ is the core tensor and $\mathbf{U} \in \mathbb{R}^{n_1 \times r_1}$, $\mathbf{V} \in \mathbb{R}^{n_2 \times r_2}$, and $\mathbf{W} \in \mathbb{R}^{n_3 \times r_3}$ are the orthogonal factor matrices that span low-dimensional subspaces along each mode. This decomposition enables a compact representation of tensor backgrounds by jointly modeling separable correlations across spatial and spectral dimensions. Based on this low-rank prior, the background component $\mathcal{B}$ can be estimated by solving the following model:

$$\min_{\mathcal{B},\mathcal{G},\mathbf{U},\mathbf{V},\mathbf{W}} \quad \|\mathcal{Y} - \mathcal{B} - \mathcal{E}\|_F^2 + \lambda\psi(\mathcal{B})$$
$$\text{s.t.} \quad \mathcal{B} = \mathcal{G} \times_1 \mathbf{U} \times_2 \mathbf{V} \times_3 \mathbf{W} \quad (7)$$

where $\mathcal{Y}$ is the observed tensor, $\mathcal{E}$ represents the sparse noise or foreground, $\psi(\cdot)$ denotes a background regularization, and $\lambda > 0$ is a balancing parameter.

Traditional regularization terms, such as total variation (TV) regularizer, typically assume uniform weighting across all spatial locations, thereby implying a homogeneous degree of continuity throughout the HSI. However, as shown in Fig. 1(c), the column vectors of the factor matrices obtained from Tucker decomposition exhibit varying levels of smoothness, revealing significant heterogeneity in spatial continuity. This discrepancy indicates the limitations of fixed-weight regularization, which struggles to characterize diverse structural patterns in real-world scenes. Motivated by this observation, we propose a Tucker decomposition-based deep prior subspace model, as illustrated in Fig. 1(a), to more effectively encode adaptive spatial smoothness and capture the intrinsic background priors of HSI data.

C. Prior Characterization

The background tensor in HSI typically exhibits a low-rank structure, which can be effectively characterized using Tucker decomposition. As shown in Fig. 1(b), the resulting core tensor contains predominantly near-zero values, where darker elements indicate larger magnitudes. Statistical analysis further confirms that the energy of the core tensor is highly concentrated around zero, implying inherent sparsity. Therefore, incorporating a sparsity constraint on the core tensor can enhance the compactness of the low-rank background representation.

In addition to low-rankness, local spatial smoothness serves as another essential prior for background modeling in HSI. Fig. 1(c) demonstrates that the spatial smoothness of the original data can be faithfully transferred to the continuity of the column vectors in the Tucker factor matrices $\mathbf{U}$, $\mathbf{V}$, and $\mathbf{W}$. To exploit this property, we introduce INRs to parameterize these factor matrices. Since INRs represent signals as continuous coordinate-based functions, their outputs naturally preserve spatial continuity. Moreover, their learnable nature allows the continuity level to be adaptively adjusted according to the underlying data, enabling flexible modeling of spatial smoothness priors. By jointly leveraging the sparsity property of the core tensor and the continuity characteristics of INR-parameterized factor matrices, our model effectively captures complementary low-rank and smoothness priors, thereby achieving more accurate and robust background representation.

Considering the above prior analysis of the background tensor, we introduce low-rank tensor function representation (LRTFR) framework used in [49] to represent the background of HSI. According to the Tucker decomposition framework, a third-order tensor $\mathcal{X} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ can be decomposed into a core tensor $\mathcal{G} \in \mathbb{R}^{r_1 \times r_2 \times r_3}$ and three factor matrices

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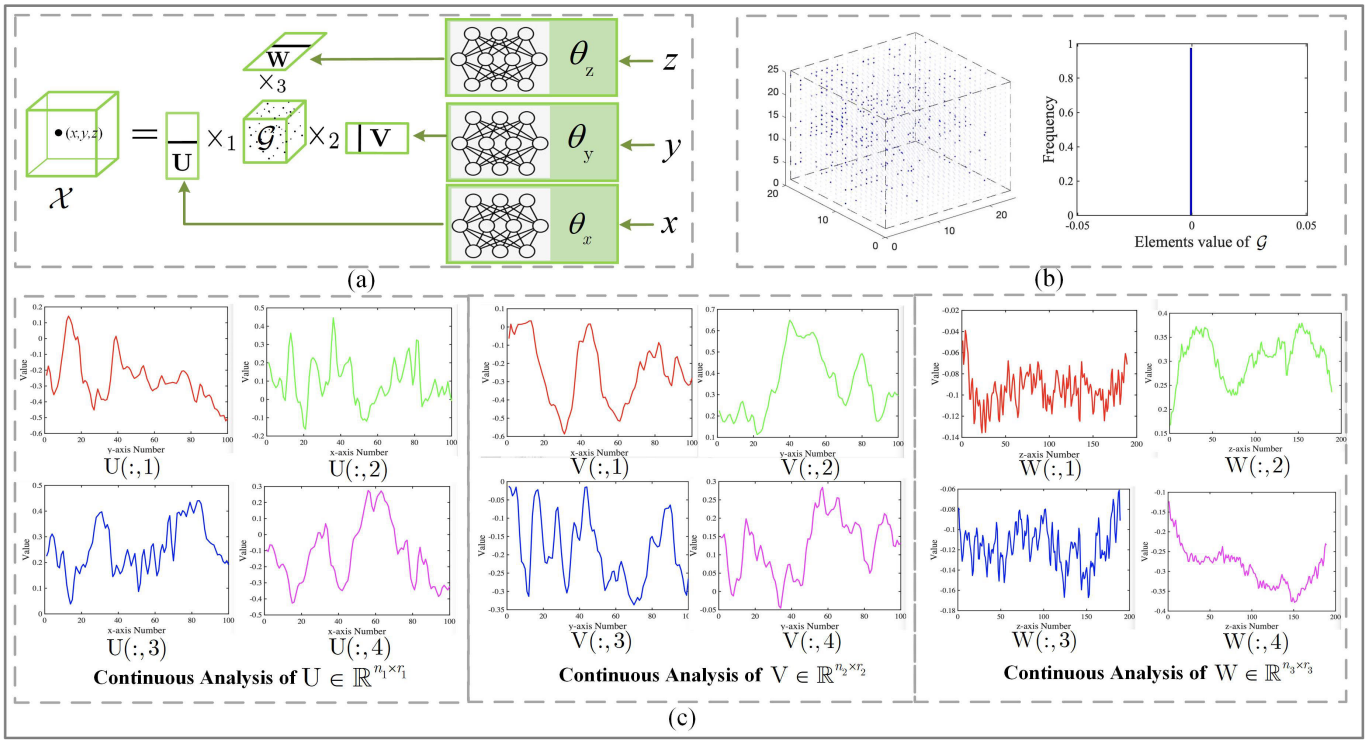


Fig. 1. Illustration of the proposed ITSR-ℓ_1 model. (a) ITSR for continuous third-order background tensor representation. (b) Left panel shows the elements in the core tensor $\mathcal{G}$ of background $\mathcal{B}$, where higher color intensity represents larger entries within the core tensor. The right panel presents the frequency histogram illustrating the distribution of element values within the core tensor $\mathcal{G}$. (c) Detailed analysis of the continuity of column vectors within the factor matrices $\mathbf{U}$, $\mathbf{V}$, and $\mathbf{W}$.

$\mathbf{U}_i \in \mathbb{R}^{r_i \times n_i}, i = 1, 2, 3$. Then, the value at any value at coordinate (i, j, k) can be expressed as

$$\boldsymbol{\mathcal{X}}(i, j, k) = \boldsymbol{\mathcal{G}} \times_1 \mathbf{U}_1(:, i) \times_2 \mathbf{U}_2(:, j) \times_3 \mathbf{U}_3(:, k). \quad (8)$$

To enhance the Tucker decomposition's ability to capture complex and nonlinear background structures in real-world data, Luo et al. [49] introduced the three bounded factor functions $f_x(\cdot)$ with the input coordinate $x_f := [1, \ldots, n_1]$, $f_y(\cdot)$ with the input coordinate $\mathbf{y}_f := [1, \ldots, n_2]$, and $f_z(\cdot)$ with the input coordinate $\mathbf{z}_f := [1, \ldots, n_3]$ to represent three matrices $\{\mathbf{U}_i\}_{i=1}^3$, respectively. Then, Luo et al. [49] give the definition of the LRTFR framework with F-rank of $[r_1, r_2, r_3]$(*i.e.*, $\boldsymbol{\mathcal{G}} \in \mathbb{R}^{r_1 \times r_2 \times r_3}$) as follows:

$$\boldsymbol{\mathcal{X}}(i, j, k) := f(i, j, k) = \boldsymbol{\mathcal{G}} \times_1 f_y(i) \times_2 f_y(j) \times_3 f_z(k). \quad (9)$$

Here, we used fully connected network with a sine activation function to represent factor function. For example, the factor function $f_x(\cdot)$ can be rewritten as

$$f_{\theta_x}(i) := \mathbf{H}_d(\sigma(\mathbf{H}_{d-1} \cdots \sigma(\mathbf{H}_1 i))) \in \mathbb{R}^{r_1} \quad (10)$$

with the learnable parameters $\theta_x = \{\mathbf{H}_l\}_{l=1}^d$, where $\sigma(\cdot)$ denotes a sine activation function, and i is index in set $\mathbf{x}_f$. Then, each column of $\mathbf{U}_1(:, i)$ can be represent by $f_x(i, \theta_x)$. By noting the network parameters of $f_y(\cdot)$ and $f_z(\cdot)$ as θ_y and θ_z, respectively, the overall representation framework can be formulated as

$$\begin{aligned} \boldsymbol{\mathcal{X}}(i, j, k) &:= f(i, j, k) := \left[\boldsymbol{\mathcal{G}}; f_{\theta_x}; f_{\theta_y}; f_{\theta_z}\right](i, j, k) \\ &:= \boldsymbol{\mathcal{G}} \times_1 f_{\theta_x}(i) \times_2 f_{\theta_y}(j) \times_3 f_{\theta_z}(k) \end{aligned} \quad (11)$$

TABLE I
SUMMARY OF PRIOR CHARACTERIZATION

Prior	Technique
Global low-rank property of $\mathcal{B}$	Tucker decomposition $[\boldsymbol{\mathcal{G}}; \mathbf{U}_1, \mathbf{U}_2, \mathbf{U}_3]$
Local-smooth of $\mathcal{B}$	TV regularizer, i.e., $\|\mathcal{B}\|_{TV}$
Local-smooth of factor $\{\mathbf{U}_i\}_{i=1}^3$	INR representation $\mathbf{U}_i := f_\theta(\cdot)$
Sparsity of Core tensor $\mathcal{G}$	$\|\mathcal{G}\|_1$ regularizer
Sparsity of anomaly tensor $\mathcal{E}$	$\|\mathcal{E}\|_1$ regularizer

where $\Theta = \{\theta_x, \theta_y, \theta_z\}$ denotes the set of learnable parameters to be updated.

Parameterizing the factor matrices with an INR captures the intrinsic continuity of their bases. However, decomposing an image into a combination of 1-D bases may be insufficient for accurately modeling the smoothness of its 2-D spatial structure. Therefore, incorporating TV regularization can further encode the local smoothness of the entire image, which is also a widely adopted technique in the existing studies [30], [50].

D. Proposed Method

According to the above analysis, the identified priors and the corresponding techniques employed to exploit them are summarized in Table I.

According to Table I, the background model integrates multiple structural priors through corresponding techniques.

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First, the global low-rank prior is enforced by representing the background as a low-rank tensor function, achieved by combining Tucker decomposition with INR-parameterized factor matrices

$$\begin{aligned}\boldsymbol{\mathcal{B}}(i,j,k) &:= f(i,j,k) := \left[\boldsymbol{\mathcal{G}}; f_{\theta_x}, f_{\theta_y}, f_{\theta_z}\right](i,j,k) \\ &:= \boldsymbol{\mathcal{G}} \times_1 f_{\theta_x}(i) \times_2 f_{\theta_y}(j) \times_3 f_{\theta_z}(k) \end{aligned} \tag{12}$$

where the multilinear ranks F-rank$[f] = (r_1, r_2, r_3)$ characterize the intrinsic dimensionality of the above tensor function [49]. The Tucker core $\boldsymbol{\mathcal{G}}$ is further constrained by an ℓ_1 regularizer to strengthen its inherent sparsity and thus enhance the global low-rank structure.

In addition to low rankness, smoothness constitutes another key prior of the background. As summarized in the table, smoothness is incorporated at two levels: 1) spatial local smoothness of $\boldsymbol{\mathcal{B}}$ through an STV regularizer and 2) continuous local smoothness of the factor matrices through INR representations. Together, these components produce an implicit regularization mechanism that couples global low-rankness with multidimensional smoothness. Noise is assumed to be Gaussian, while anomalies are sparse; thus, an ℓ_1 penalty on $\boldsymbol{\mathcal{E}}$ is introduced to encode the sparsity prior of anomalies. Finally, the resulting anomaly detection model is given as follows:

$$\begin{aligned} \min_{\boldsymbol{\mathcal{G}},\boldsymbol{\mathcal{E}},\theta_x,\theta_y,\theta_z} \quad & \|\boldsymbol{\mathcal{Y}} - \boldsymbol{\mathcal{B}} - \boldsymbol{\mathcal{E}}\|_F^2 + \lambda\|\boldsymbol{\mathcal{B}}\|_{\mathrm{STV}} + \beta\|\boldsymbol{\mathcal{G}}\|_1 + \gamma\|\boldsymbol{\mathcal{E}}\|_1 \\ \text{s.t.} \quad & \boldsymbol{\mathcal{B}} = \left[\boldsymbol{\mathcal{G}}; \mathrm{f}_{\theta_x}, \mathrm{f}_{\theta_y}, \mathrm{f}_{\theta_z}\right] \end{aligned} \tag{13}$$

where $\|\cdot\|_F^2$, $\|\cdot\|_1$, and $\|\cdot\|_{\mathrm{STV}}$ denote the ℓ_2-norm, ℓ_1-norm, and STV regularizer, respectively.

We utilize the alternating minimization algorithm to tackle the model. Specifically, we tackle the following subproblems in the tth iteration:

$$\begin{cases} \boldsymbol{\mathcal{B}}^{t+1} & := \min_{\boldsymbol{\mathcal{G}},\theta_x,\theta_y,\theta_z} \|\boldsymbol{\mathcal{Y}} - \boldsymbol{\mathcal{B}} - \boldsymbol{\mathcal{E}}^t\|_F^2 + \lambda\|\boldsymbol{\mathcal{B}}\|_{\mathrm{STV}} + \beta\|\boldsymbol{\mathcal{G}}\|_1 \\ \boldsymbol{\mathcal{E}}^{k+1} & := \min_{\boldsymbol{\mathcal{E}}} \|\boldsymbol{\mathcal{Y}} - \boldsymbol{\mathcal{B}}^{t+1} - \boldsymbol{\mathcal{E}}\|_F^2 + \gamma\|\boldsymbol{\mathcal{E}}\|_1. \end{cases} \tag{14}$$

1) Update $\boldsymbol{\mathcal{B}}$: Let $\boldsymbol{\mathcal{L}}(\boldsymbol{\mathcal{G}}, \theta_x, \theta_y, \theta_z) = \|\boldsymbol{\mathcal{Y}} - \boldsymbol{\mathcal{B}} - \boldsymbol{\mathcal{E}}\|_F^2 + \lambda\|\boldsymbol{\mathcal{B}}\|_{\mathrm{STV}} + \beta\|\boldsymbol{\mathcal{G}}\|_1$; first, we employ one step of the Adam algorithm to update $\{\boldsymbol{\mathcal{G}}, \theta_x, \theta_y, \theta_z\}$

$$\begin{cases} \theta_x^{t+1} := \theta_x^t - \alpha\partial\boldsymbol{\mathcal{L}}\left(\boldsymbol{\mathcal{G}}, \theta_x, \theta_y, \theta_z\right)/\partial\theta_x \\ \theta_y^{t+1} := \theta_y^t - \alpha\partial\boldsymbol{\mathcal{L}}\left(\boldsymbol{\mathcal{G}}, \theta_x, \theta_y, \theta_z\right)/\partial\theta_y \\ \theta_z^{t+1} := \theta_z^t - \alpha\partial\boldsymbol{\mathcal{L}}\left(\boldsymbol{\mathcal{G}}, \theta_x, \theta_y, \theta_z\right)/\partial\theta_z \\ \boldsymbol{\mathcal{G}}^{t+1} := \boldsymbol{\mathcal{G}}^t - \alpha\partial\boldsymbol{\mathcal{L}}\left(\boldsymbol{\mathcal{G}}, \theta_x, \theta_y, \theta_z\right)/\partial\boldsymbol{\mathcal{G}} \\ \boldsymbol{\mathcal{B}}^{t+1} := \left[\boldsymbol{\mathcal{G}}^{t+1}; f_{\theta_x}, f_{\theta_y}, f_{\theta_z}\right] \end{cases} \tag{15}$$

then, we obtain the reconstructed $\boldsymbol{\mathcal{B}}^{t+1}$.

2) Update $\boldsymbol{\mathcal{E}}^{t+1}$: The closed-form solution of subproblem about $\boldsymbol{\mathcal{E}}$ can be obtained via the elementwise soft-thresholding operator [51]

$$\boldsymbol{\mathcal{E}}^{t+1} = \mathcal{S}_{1/\gamma}\left(\boldsymbol{\mathcal{Y}}^t - \boldsymbol{\mathcal{B}}^t\right). \tag{16}$$

The complete procedure is presented in Algorithm 1.

Algorithm 1 ITSR-ℓ_1 Algorithm

1: **Input:** HSI tensor $\boldsymbol{\mathcal{Y}}$; maximum iteration number $K = 5000$; learning rate $\alpha = 1e^{-4}$; background tensor function rank F-rank$[f] = (r_1, r_2, r_3)$.
2: **Initialize:** $k = 1$ and all variables.
3: **for** $k = 1 : K$ **do**
4: Update $\theta_x, \theta_y, \theta_z$ and $\boldsymbol{\mathcal{G}}$ via Eqs. (15).
5: Update $\boldsymbol{\mathcal{E}}$ via (16).
6: **end for**
7: **Output:** anomaly tensor $\boldsymbol{\mathcal{E}}$.

E. Computational Complexity Analysis

Assume that the observed tensor $\boldsymbol{\mathcal{Y}}$ has dimensions $n_1 \times n_2 \times n_3$, and its Tucker rank is given by (r_1, r_2, r_3). Let $N = n_1 n_2 n_3$ denote the number of entries in $\boldsymbol{\mathcal{Y}}$, $R = r_1 r_2 r_3$ the size of the core tensor, and let each INR network contain l layers with m neurons per layer. In each iteration of the alternating minimization, the major computational cost consists of the following parts.

1) *INR Forward/Backward Computation:* Generating the factor matrices via three coordinate-based INRs, its time computational cost is $\mathcal{O}\left(lm^2\sum_{i=1}^3 n_i\right)$.
2) *Tucker Decomposition:* Given a core tensor $\boldsymbol{\mathcal{G}} \in \mathbb{R}^{r_1\times r_2\times r_3}$ and three factor matrices $\mathbf{U} \in \mathbb{R}^{n_1\times r_1}$, $\mathbf{V} \in \mathbb{R}^{n_2\times r_2}$, and $\mathbf{W} \in \mathbb{R}^{n_3\times r_3}$, the Tucker reconstruction $\boldsymbol{\mathcal{B}} = \boldsymbol{\mathcal{G}} \times_1 \mathbf{U} \times_2 \mathbf{V} \times_3 \mathbf{W}$ can be computed by three successive mode-n products. The computational cost of each step is $\mathcal{O}\left(R\sum_{i=1}^3 n_i\right)$.
3) *Loss and Regularization:* Computing the data fidelity, STV regularizer, and ℓ_1 terms costs $\mathcal{O}(N + R)$.
4) *Updating $\boldsymbol{\mathcal{E}}$:* The soft-thresholding step requires $\mathcal{O}(N)$.

Combining the above, the overall per-iteration time complexity is $\mathcal{O}\left(2N + R + (R + lm^2)\sum_{i=1}^3 n_i\right)$.

III. Experimental Results and Discussion

To demonstrate the effectiveness of our method, we compare the proposed ITSR-ℓ_1 with several representative approaches, including RX [52], RPCA [53], GTVLRR [23], E-3DTV [50], PTA [54], TRPCA [55], LRTDTV [27], as well as an ablation version referred to as ITSR. In Table II, the abbreviation DL refers to deep learning methods, while NDL indicates nondeep learning methods. All experiments were conducted on a computing device running Python 3.7 and PyTorch 1.10, equipped with a 6-GB NVIDIA GeForce RTX 2060 GPU.

A. Datasets

In the experiments, we utilize the following four real datasets that are commonly employed in peer HAD.

1) *Airborne Visible Infrared Imaging Spectrometer (AVIRIS) Airplane Datasets:* The AVIRIS-1 dataset, derived from the AVIRIS image collected over San Diego, is positioned in the image's top-left corner and has a size of $100 \times 100 \times 189$, as illustrated in Fig. 2(a1). This region contains three airplanes regarded

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TABLE II
QUANTITATIVE RESULTS OVER DIFFERENT HAD ALGORITHMS ON FOUR DATASETS. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Datasets	AUC Values	Matrix-based operations					Tensor-based operations				
		NDL				DL	NDL			DL	
		RX	RPCA	GTVLRR	E-3DTV	GVAE	PTA	TRPCA	LRTDTV	ITSR	ITSR-ℓ_1
AVIRIS-1	$AUC_{(P_D,P_F)}$	0.9768	0.9591	0.9685	0.9659	0.9773	0.9221	0.9029	0.9957	0.9923	0.9938
	$AUC_{(P_D,\tau)}$	0.0926	0.0585	0.2328	0.3086	0.5772	0.3423	0.4249	0.3152	0.5197	0.5674
	$AUC_{(P_F,\tau)}$	0.0172	0.0255	0.0574	0.0608	0.1080	0.1278	0.1187	0.0472	0.0785	0.0934
	$AUC_{(OD)}$	1.0392	0.9265	1.1337	1.2312	1.4568	1.1654	1.2934	1.2523	1.4335	1.4677
Urban-1	$AUC_{(P_D,P_F)}$	0.9934	0.9916	0.9093	0.9794	0.9852	0.9778	0.9823	0.9870	0.9934	0.9959
	$AUC_{(P_D,\tau)}$	0.3442	0.3117	0.4403	0.6078	0.6130	0.3729	0.6956	0.5432	0.3442	0.5256
	$AUC_{(P_F,\tau)}$	0.0329	0.0496	0.1261	0.2374	0.1869	0.0925	0.2450	0.0868	0.0329	0.0468
	$AUC_{(OD)}$	1.3047	1.2537	1.2235	1.3498	1.4113	1.2582	1.4329	1.4434	1.3047	1.4747
Urban-2	$AUC_{(P_D,P_F)}$	0.9946	0.9960	0.9967	0.9382	0.9992	0.9828	0.9456	0.9904	0.9979	0.9993
	$AUC_{(P_D,\tau)}$	0.2880	0.2347	0.2295	0.8871	0.3213	0.1400	0.6048	0.2668	0.2592	0.2903
	$AUC_{(P_F,\tau)}$	0.0611	0.0283	0.0487	0.5681	0.0672	0.1976	0.0325	0.0459	0.0194	0.0305
	$AUC_{(OD)}$	1.2215	1.2024	1.1775	1.2572	1.2533	0.9252	1.5179	1.2113	1.2377	1.2591
Segundo	$AUC_{(P_D,P_F)}$	0.8249	0.7953	0.8168	0.8982	0.6111	0.8249	0.7474	0.8936	0.9930	0.9952
	$AUC_{(P_D,\tau)}$	0.0036	0.0228	0.1580	0.3614	0.4041	0.0875	0.0558	0.0664	0.1169	0.1424
	$AUC_{(P_F,\tau)}$	0.0022	0.0132	0.0847	0.2009	0.3692	0.0506	0.0348	0.0160	0.0131	0.0199
	$AUC_{(OD)}$	0.8262	0.8049	0.8901	1.0587	0.6460	0.8617	0.7684	0.9440	1.0968	1.1178

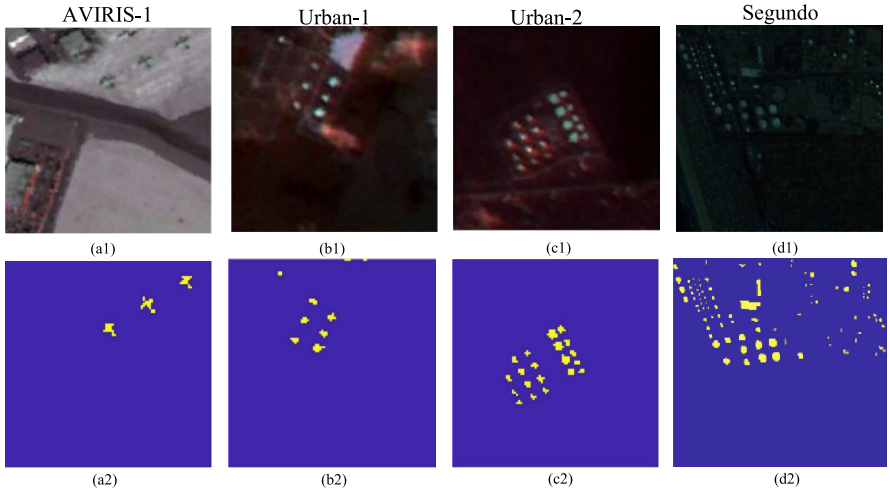


Fig. 2. Four datasets used in our experiments are AVIRIS-1, Urban-1, Urban-2, and Segundo. Displayed in the first row are the false-color images of the four scenes (a1)–(d1), and in the second row are the corresponding ground-truth maps (a2)–(d2).

as anomalies, and their ground truth can be seen in Fig. 2(a2).

2) *Urban (ABU) Datasets:* The urban dataset, acquired by the AVIRIS sensor, comprises four different scenes. In this work, we focus on two of them, namely, Urban-1 and Urban-2. Urban-1 has a spatial-spectral size of $100 \times 100 \times 204$, and its false-color image and manually labeled ground truth are displayed in Fig. 2(b1) and (b2), respectively. Urban-2, with dimensions of $100 \times 100 \times 207$, is similarly illustrated through its false-color representation and ground truth in Fig. 2(c1) and (c2).

3) *Segundo Dataset:* The fourth dataset, collected by the AVIRIS sensor over El Segundo, CA, USA, covers an area of 250×300 pixels with 224 spectral bands ranging from 366 to 2496 nm. As shown in Fig. 2(d1), the oil refinery structures, such as storage tanks and towers, are the designated targets. The ground truth is presented in Fig. 2(d2).

B. Evaluation Metrics and Parameter Settings

1) Anomaly Detection Evaluation Metrics: We adopt the 3-D receiver operating characteristic (3D-ROC) curve, represented as (P_D, P_F, τ) [56] to comprehensively assess the performance of the compared algorithms. Three 2D-ROC curves—(P_D, P_F), (P_D, τ), and (P_F, τ)—are employed for performance evaluation, with their respective AUC values denoted as $AUC_{(P_D,P_F)}$, $AUC_{(P_D,\tau)}$, and $AUC_{(P_F,\tau)}$. These metrics correspond to comprehensive detection accuracy, anomaly

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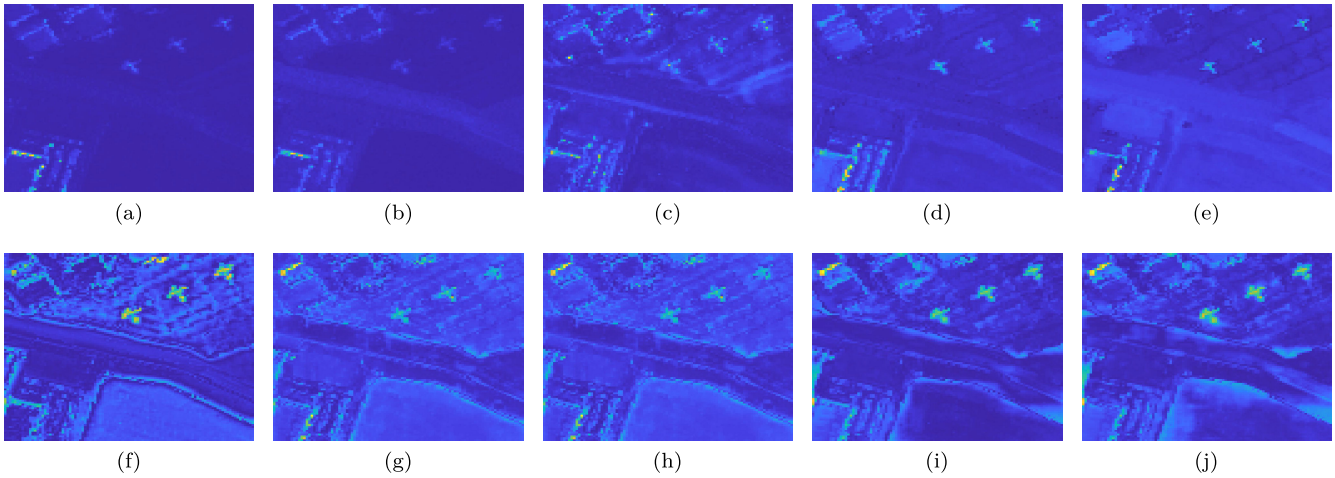


Fig. 3. 2-D detection result plots of the compared algorithms on the AVIRIS-1 dataset. (a) RX. (b) RPCA. (c) GTVLRR. (d) E-3DTV. (e) GVAE. (f) PTA. (g) TRPCA. (h) LRTDTV. (i) ITSR. (j) ITSR-ℓ_1.

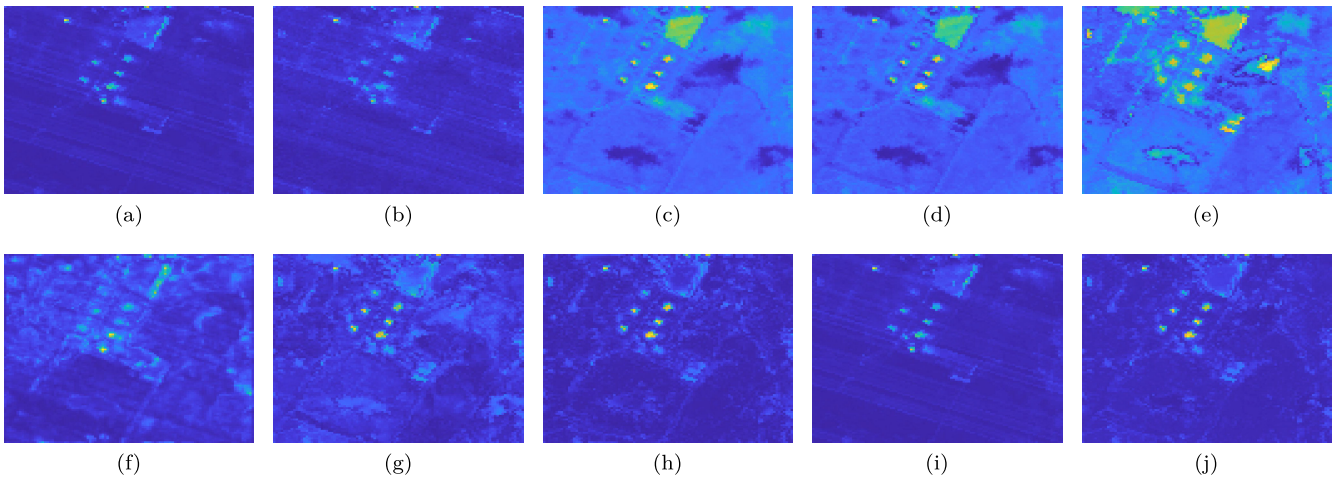


Fig. 4. 2-D detection result plots of the compared algorithms on Urban-1 dataset. (a) RX. (b) RPCA. (c) GTVLRR. (d) E-3DTV. (e) GVAE. (f) PTA. (g) TRPCA. (h) LRTDTV. (i) ITSR. (j) ITSR-ℓ_1.

preservation, and background suppression performance. Moreover, we introduce a new metric, $\mathrm{AUC}_{(\mathrm{OD})}$, to evaluate its overall performance, which is defined as follows:

$$\mathrm{AUC}_{(\mathrm{OD})} = \mathrm{AUC}_{(\mathrm{P_D},\mathrm{P_F})} + \mathrm{AUC}_{(\mathrm{P_D},\tau)} - \mathrm{AUC}_{(\mathrm{P_F},\tau)}. \tag{17}$$

A desirable detection performance is indicated when $\mathrm{AUC}_{(\mathrm{P_D},\mathrm{P_F})}$ and $\mathrm{AUC}_{(\mathrm{P_D},\tau)}$ approach 1, $\mathrm{AUC}_{(\mathrm{P_F},\tau)}$ approaches 0, and $\mathrm{AUC}_{(\mathrm{OD})}$ approaches 2. Furthermore, it is expected that the corresponding curves of $(\mathrm{P_D}, \mathrm{P_F})$, $(\mathrm{P_D}, \tau)$, and $(\mathrm{P_F}, \tau)$ are located near the upper left, upper right, and lower left corners of their respective plots.

2) Hyperparameter Settings: In model (13), λ controls the smoothness of the background, while β and γ regulate the sparsity of the core tensors for the background and anomalies, and these parameters must be carefully tuned. The core tensor is set as $\mathrm{F{-}rank}(f) = [r_1, r_2, r_3]$, where r_1 and r_2 represent the spatial dimensions of $\mathcal{G}$, which are closely related to the background's structural complexity, and r_3 denotes the spectral dimension. Owing to the strong spectral correlation in HSIs, the spectral rank is typically smaller than the spatial ranks.

C. Detection Performance

The proposed ITSR method is evaluated against nine state-of-the-art algorithms from both quantitative and qualitative perspectives. Figs. 3–6 illustrate the 2-D detection maps produced by each method across four benchmark datasets, providing a visual comparison of detection performance. In addition, Fig. 7 presents the ROC-based performance assessments, where each subfigure corresponds to one dataset and includes a 3-D ROC curve, $(\mathrm{P_D}, \mathrm{P_F}, \tau)$, along with its three 2-D projections: $(\mathrm{P_D}, \mathrm{P_F})$, $(\mathrm{P_D}, \tau)$, and $(\mathrm{P_F}, \tau)$. The boxplots in Fig. 8 present the performance distribution of the compared algorithms across all datasets, highlighting their ability to distinguish anomalies from the background. In addition to the previously discussed AUC metrics, Table II summarizes the $\mathrm{AUC}_{(\mathrm{P_D},\mathrm{P_F})}$, $\mathrm{AUC}_{(\mathrm{P_D},\tau)}$, $\mathrm{AUC}_{(\mathrm{P_F},\tau)}$, and $\mathrm{AUC}_{(\mathrm{OD})}$ values, where $\mathrm{AUC}_{(\mathrm{OD})}$ providing an overall performance comparison. Computational efficiency is provided in Table III, where the runtime of GVAE includes both training and inference phases.

We take AVIRIS-1 as an example to illustrate the proposed method's superiority over other methods. Fig. 2 presents the

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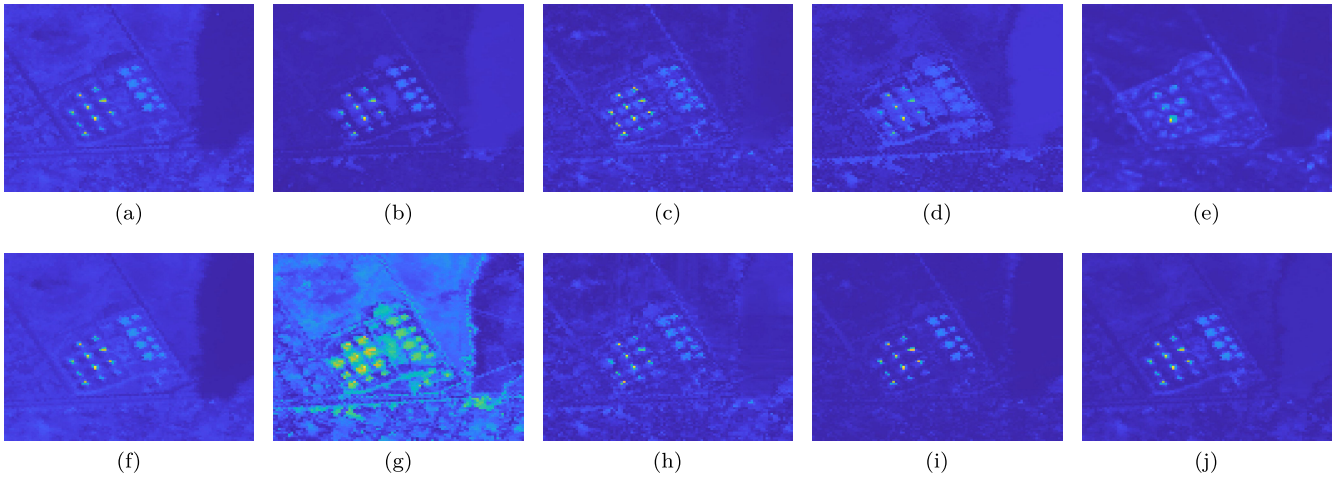


Fig. 5. 2-D detection result plots of the compared algorithms on Urban-2 dataset. (a) RX. (b) RPCA. (c) GTVLRR. (d) E-3DTV. (e) GVAE. (f) PTA. (g) TRPCA. (h) LRTDTV. (i) ITSR. (j) ITSR-ℓ_1.

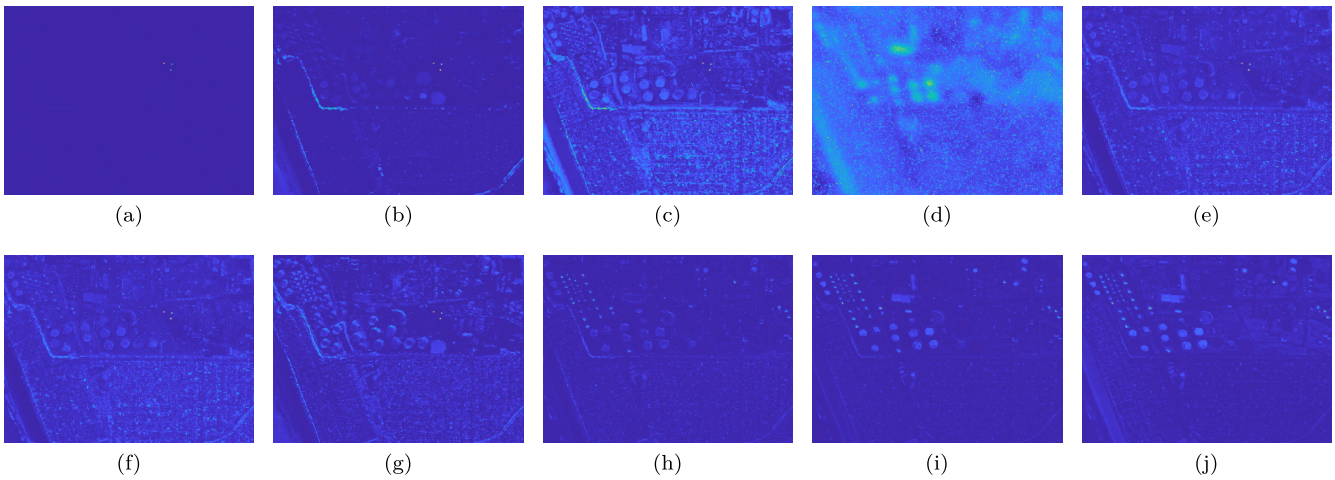


Fig. 6. 2-D detection result plots of the compared algorithms on the Segundo dataset. (a) RX. (b) RPCA. (c) GTVLRR. (d) E-3DTV. (e) GVAE. (f) PTA. (g) TRPCA. (h) LRTDTV. (i) ITSR. (j) ITSR-ℓ_1.

TABLE III

RUNTIME ANALYSIS OF DIFFERENT HAD ALGORITHMS EVALUATED ON FOUR DATASETS

Dataset	Matrix-based operations					Tensor-based operations				
	NDL				DL	NDL			DL	
	RX	RPCA	GTVLRR	E-3DTV	GVAE	PTA	TRPCA	LRTDTV	ITSR	ITSR-ℓ_1
AVIRIS-1	0.490	20.419	303.958	28.332	0.569	20.017	153.577	133.539	2.340	3.560
Urban-1	0.482	16.179	144.812	29.377	0.425	22.041	181.422	72.080	37.980	57.450
Urban-2	0.719	16.315	113.543	31.215	0.518	22.147	189.801	70.564	3.880	9.800
Segundo	10.008	87.561	648.155	48.101	0.518	461.900	1537.419	491.521	26.280	31.490

false-color images alongside the corresponding ground-truth anomaly maps for the four benchmark datasets. Based on these references, the effectiveness of the proposed method is validated through multiple forms of visual and quantitative comparisons, including the detection maps in Figs. 3–6, the ROC curves illustrated in Fig. 7, and the box-and-whisker plots depicted in Fig. 8.

1) AVIRIS-1: The ITSR-ℓ_1 achieves the best performance when the rank is set to F−rank(f) = $[25, 25, 20]$. From Fig. 3(j), it can be observed that the three anomalous airplanes are clearly detected, and the method also demonstrates satisfactory background suppression. Overall, it achieves the best comprehensive anomaly detection performance. Moreover, as shown in Fig. 7(a), the distribution of the four evaluation metric curves, (P_D, P_F), (P_D, τ), (P_F, τ), and (P_D, P_F, τ) along with the quantitative results presented in Table II, jointly verifies that the proposed method can adaptive capture the prior structural information of the background, such as low rankness and smoothness.

The LRTV, GTVLRR, E-3DTV, PTA, and the proposed ITSR-ℓ_1 apply TV regularization to the background to suppress noise and enhance anomaly contrast. As shown in Figs. 4 and 8 and Tables II and III, the SOTA deep learning method GVAE shows poor generalization, while ITSR-ℓ_1,

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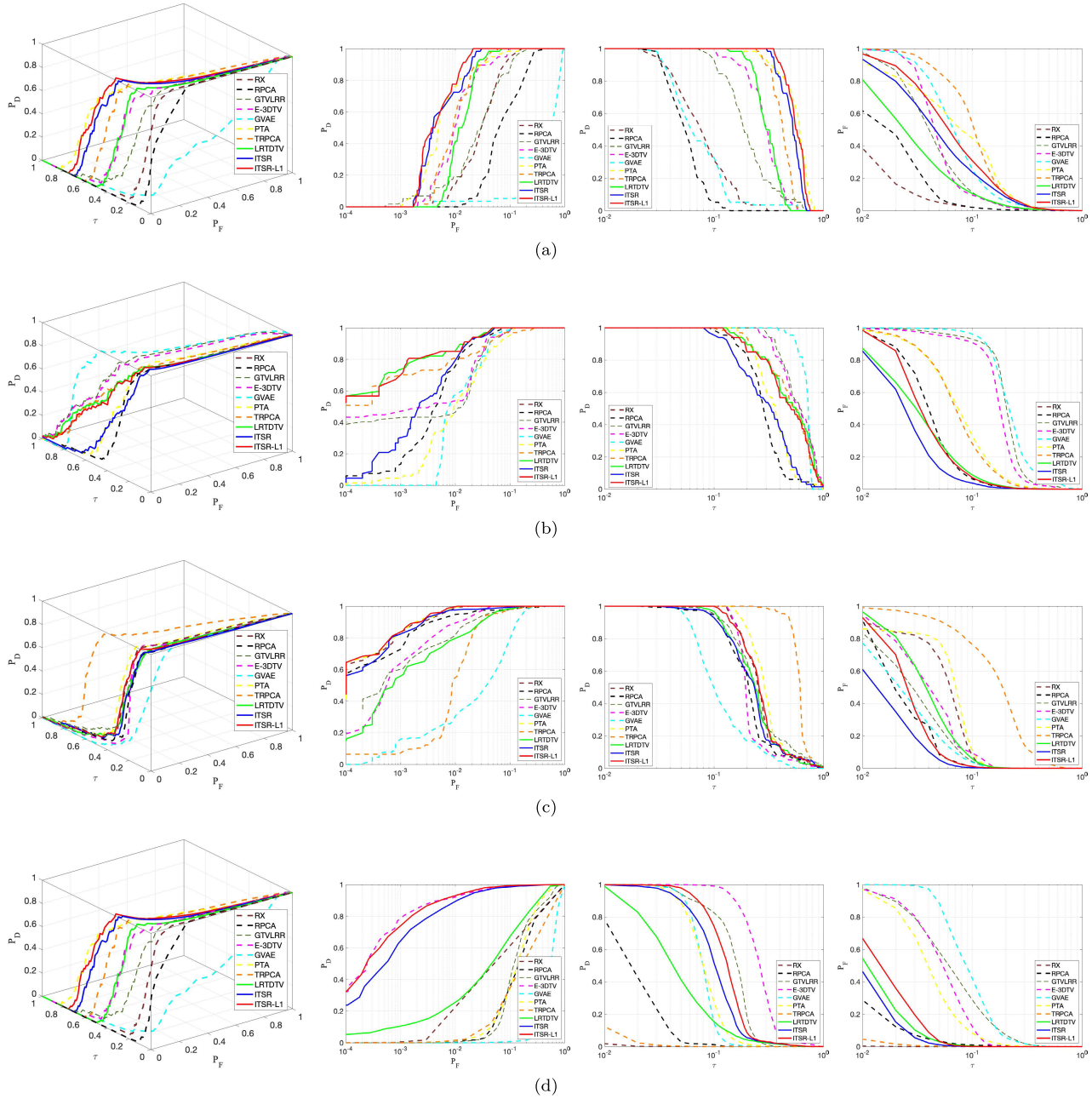


Fig. 7. ROC curves of different comparison methods on the four datasets. (a) AVIRIS-1. (b) Urban-1. (c) Urban-2. (d) Segundo. Displayed from left to right are the 3-D ROC curve, followed by the 2-D ROC curves corresponding to (P_D, P_F), (P_D, τ), and (P_F, τ), respectively.

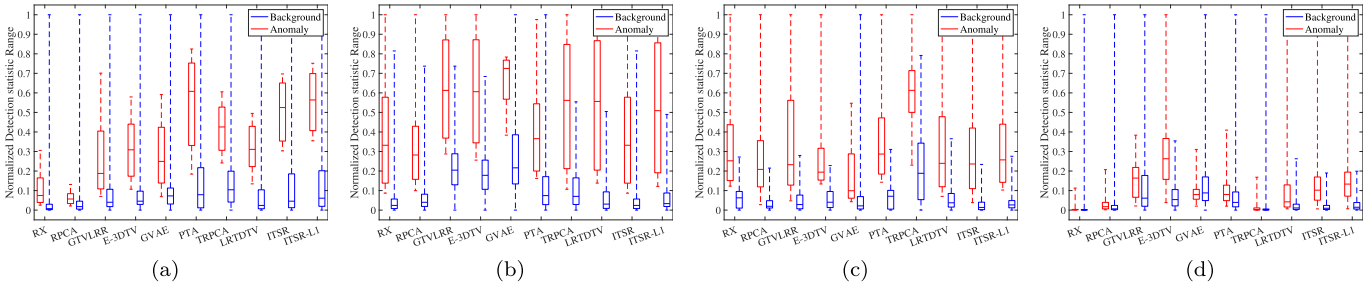


Fig. 8. Visualization of the comparative performance of different algorithms using box-and-whisker plots on four real datasets. (a) AVIRIS-1, (b) Urban-1, (c) Urban-2, and (d) Segundo.

by leveraging implicit neural networks to parameterize factor vectors, adaptively captures fine-grained continuity priors in the background. Furthermore, as depicted in Fig. 8(a), ITSR-ℓ_1 exhibits a noticeably wider interval between the red and blue boxes compared to the other nine competing methods. This indicates that ITSR-ℓ_1 achieves superior separability between

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anomalies and background, demonstrating its effectiveness in distinguishing target signals. Except for RX and GVAE, all comparison methods are based on low-rank matrix or tensor decomposition. Our proposed method employs a low-rank Tucker function representation, which enables continuous data modeling in infinite real space. As demonstrated in Figs. 3, 7, and 8 and Tables II and III, this approach more effectively captures the low-rank structure of the background, leading to more accurate reconstruction. Compared to ITSR, the proposed ITSR-ℓ_1 further introduces the ℓ_1 norm to promote background sparsity and better control low rankness. Notably, Table III shows that our method achieves lower computational cost while maintaining a high detection rate.

2) Urban-1: The ITSR-ℓ_1 achieves the best performance when the rank is set to F−rank(f) = $[33, 33, 13]$. As shown in Fig. 4(j), the anomalies are accurately identified, while the background interference is effectively suppressed. In Table II, the $\text{AUC}_{(\text{P}_\text{D},\tau)}$ of the TRPCA is higher than the proposed ITSR-ℓ_1, whereas its $\text{AUC}_{(\text{P}_\text{D},\text{P}_\text{F})}$ and $\text{AUC}_{(\text{P}_\text{F},\tau)}$ performances are inferior to the other comparison methods, which lead to a lower $\text{AUC}_{(\text{OD})}$ than those of most benchmark algorithms. As shown in Fig. 4, this indicates that TRPCA effectively preserves background suppression capability, and Fig. 7 further proves it. In summary, the analysis of both detection maps and AUC metrics verifies that ITSR-ℓ_1 exhibits superior performance in terms of overall detection accuracy, effective anomaly preservation, and background suppression.

3) Urban-2: The ITSR-ℓ_1 achieves the best performance when the rank is set to F−rank(f) = $[33, 33, 13]$. From Table II, we can see that the $\text{AUC}_{(\text{P}_\text{D},\text{P}_\text{F})}$ of the proposed ITSR-ℓ_1 is the highest. Table III shows that its time cost is also lower than that of most other comparison methods. Although TRPCA achieves the highest $\text{AUC}_{(\text{OD})}$ in Table II, its highest $\text{AUC}_{(\text{P}_\text{D},\tau)}$ particularly indicates effective foreground preservation, whereas its overall detection performance, including background suppression, is inferior to ITSR-ℓ_1, and its time consumption is relatively high. Furthermore, Tables II and III show that the ablation model ITSR exhibits the best background suppression. Compared with ITSR, the proposed ITSR-ℓ_1 performs better in foreground detection, achieves a higher $\text{AUC}_{(\text{OD})}$, and shows a larger interval between the red and blue boxes, further demonstrating its superior separation capability.

4) Segundo: The ITSR-ℓ_1 achieves the best performance when the rank is set to F−rank(f) = $[62, 75, 28]$. Through Table II and Fig. 6, the foreground detection of GVAE is the best but performs poorly in background suppression and overall detection results. GVAE utilizes the graph adjacency matrix regularization to demonstrate the relationship between the different pixels. Since the size of dataset is too large, the computer's memory space is limited, necessitating blockwise execution. This size of the Sedungo dataset makes the anomaly detection challenging to capture the complex features in the background accurately; however, our proposed ITSR-ℓ_1 can effectively capturing the background complex prior information and achieving accurate anomaly detection. Moreover, Table II and Figs. 6 and 7 demonstrate the superiority of our proposed method compared to other algorithms.

IV. Conclusion

In this article, we proposed a novel HAD method, termed ITSR-ℓ_1. This by employing Tucker decomposition to explicitly capture the intrinsic low-rank structure of the background, exploiting the sparsity of the core tensor to further enhance the modeling of the low-rank background. Moreover, the local smoothness inherent in the background is mapped to the continuous structure of the column vectors within the factor matrices obtained by Tucker decomposition. Through implicit parameterization of neural networks, ITSR-ℓ_1 adaptively captures detailed continuity structures within the background. The results of four datasets show that the proposed method delivers better performance than the existing methods.

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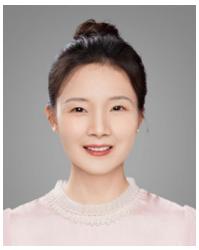
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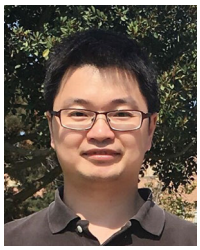
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